Kousha Ghodsizad

Systems & Infrastructure Engineer

CONTACT

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TECHNICAL SKILLS

Languages & Backend
Go, Python, PHP, Shell, APIs, gRPC, GraphQL, services, workers, CLIs

Containers & Orchestration
Kubernetes, K3s, Docker, Helm, Cilium, Flannel, service mesh

Delivery & Workflows
Argo CD, Argo Rollouts, Argo Workflows, Argo Events, Temporal, GitLab CI, Jenkins, Harbor, Nexus

Infrastructure & Automation
AWS, private cloud, OpenStack, Terraform, Puppet, HashiCorp Vault, HashiCorp Consul

Data & Messaging
MySQL, PostgreSQL, Redis, RabbitMQ, Kafka, NATS, Cassandra, ClickHouse, Elasticsearch, MongoDB

Observability & Reliability
Prometheus, Grafana, metrics, logging, operational dashboards, readiness checks, rollout safety

Architecture
System design, reliability engineering, automation strategy, modular architecture, domain boundaries, technical review, engineering standards

EDUCATION

BSc Computer Software Engineering
Islamic Azad University · 2011 — 2015

ASc Computer Programming
Islamic Azad University · 2008 — 2011

References available on request.

SUMMARY

Systems & Infrastructure / Platform Engineer with a backend engineering background. I build and operate production platforms: Kubernetes-based infrastructure, CI/CD and workflow systems, and the developer tooling teams use to ship and run their services. My work covers infrastructure automation, observability, reliability, and distributed systems, with a focus on building platforms that are secure, resilient, and straightforward to operate.

I take ownership of the systems I work on, from design and implementation through production operation and ongoing improvement. I enjoy hard technical problems and the architectural decisions that come with them, and I build tools that let engineering teams work independently. I hold a high bar for engineering quality and for systems that remain reliable and maintainable as they grow.

CORE EXPERTISE

Platform Engineering & Developer Tooling: Internal platforms, CLIs, self-service workflows, secure execution. Go-based platform tooling, including CLIs and plugin SDKs, with plugin management and access-controlled execution on Linux. Workflow automation with Argo Workflows, Argo Events, and Temporal, and self-service tooling that lets teams operate independently.

Kubernetes & Application Delivery: Kubernetes infrastructure, GitOps, progressive delivery, CI/CD. Kubernetes-based production infrastructure and application delivery with Helm, Argo CD for GitOps, and Argo Rollouts for progressive delivery in private cloud environments. CI/CD with GitLab CI and Jenkins, deployment and merge-request automation, artifact workflows, and release guardrails.

Infrastructure as Code & Secrets: Provisioning, secrets management, service discovery. Terraform, HashiCorp Vault, and HashiCorp Consul used alongside Kubernetes to provision infrastructure, manage secrets, and register services.

High Availability & Reliability: Resilient infrastructure, rollout safety, operational readiness. Highly available backend systems across multiple data centres, with request routing, workload distribution, and failover that keep them operating under heavy load and through data-centre loss. Deployment hardening, safe rollout patterns, readiness checks, and the operational groundwork that lets production changes ship with confidence.

Observability & Production Operations: Metrics, monitoring, debugging, incident investigation. Prometheus and Grafana, metrics, logs, dashboards, and post-deployment feedback loops for production systems. Investigating and resolving production issues across application, infrastructure, pipeline, and database layers.

Backend & Distributed Systems: Go services, APIs, distributed systems, messaging. Backend services, APIs, workers, CLIs, service boundaries, integration layers, and data flows, with particular strength in Go. Practical experience with messaging and data systems such as Redis, RabbitMQ, Kafka, NATS, MySQL, PostgreSQL, Cassandra, ClickHouse, Elasticsearch, and MongoDB.

Architecture & Technical Leadership: System design, technical decisions, standards. Architecture and service-boundary design, platform decisions, engineering standards, code review, and mentoring. Career progression from backend engineer to team lead, head of development, and infrastructure/platform engineer, with growing responsibility for system design and production systems.

EXPERIENCE

Adyen

Jul 2023 — Present

Infrastructure Engineer — CI/CD & Platform · Amsterdam, Netherlands

- Designed and built adyencli in Go, the CLI of a plugin management platform: teams develop plugins with Adyen-provided SDKs in four programming languages, submit them to be built and distributed through Adyen's package-management infrastructure, and the platform manages their lifecycle and controlled execution in Linux environments.
- Built access control and controlled, secure plugin execution into the platform, and worked on its backend services and supporting infrastructure on Kubernetes, Helm, Argo CD, HashiCorp Vault, and HashiCorp Consul.
- Designed the backend architecture of a separate highly available system distributed across four geographically separated data centres, routing requests to suitable, fast-responding services, with the architecture built to handle heavy workloads and keep operating when one or more data centres become unavailable.
- Built a solution for operating Argo Workflows across multiple Kubernetes clusters and data centres, including the workflow controllers and their supporting database and state infrastructure, with execution responsibility coordinated across data centres that all had access to the workflow workload.
- Used Kubernetes Leader Election as part of determining which controller owned a workflow's execution. Workflow and step state were recorded in the database, so when a lost node left a step marked Failed and ownership moved to another controller, the new owner retried that step from the recorded state and the workflow continued from there instead of restarting; surviving data centres took over when a data centre failed.
- Contributed to a modular, self-service automation platform in Python: teams configure repository-specific automation by writing Gherkin scenarios that invoke reusable modules, instead of implementing each workflow themselves. It uses NATS as its event and message infrastructure and provides detailed Prometheus and Grafana monitoring of automation runs, down to individual merge requests and their pipeline executions.
- Across this work, used Jenkins and GitLab CI for CI/CD pipelines, Argo CD and Argo Rollouts for deployment and progressive delivery, Argo Events for event-driven automation, and Temporal for workflow orchestration.
- Used Terraform for infrastructure provisioning, Puppet for configuration management, HashiCorp Vault for secrets management, and HashiCorp Consul for service discovery, and worked with Flannel networking and service mesh technology in K3s environments.

Go · Python · Gherkin · Kubernetes · K3s · Helm · Argo CD · Argo Workflows · Argo Events · Argo Rollouts · Temporal · Jenkins · GitLab CI · Terraform · Puppet · HashiCorp Vault · HashiCorp Consul · Flannel · NATS · Redis · Prometheus · Grafana · Linux

Studocu

Jul 2022 — Jul 2023

Senior Software Engineer · Amsterdam, Netherlands

- Improved backend and platform-facing areas around website performance, crawler accessibility, and production reliability.
- Worked with operations and platform teams to add monitoring across local, staging, and production environments.
- Built CI/CD checks for SEO and performance testing to make release quality more repeatable and easier to validate.
- Investigated bottlenecks across application and database layers and translated findings into practical engineering improvements.

Amazon Web Services (AWS) · S3 · SNS · SQS · RDS · DevOps · PHP · Laravel · Go · Big Data · Test Driven Development (TDD) · MySQL · ElasticSearch · Redis

KolayIK

May 2021 — May 2022

Senior Software Engineer / Backend Team Lead · Istanbul, Turkey

- Led backend and platform-oriented work across HR systems, expense management, data import/export, and queue-based services.
- Applied modular architecture, domain boundaries, TDD, messaging, AWS services, and Kubernetes where appropriate to improve maintainability and operational behaviour.
- Built automation for failure reporting and operational feedback, including queue handling and Slack-integrated reporting.
- Supported delivery through technical review, design direction, mentoring, and pragmatic engineering standards.

Wingie Enuygun Group

May 2020 — Apr 2021

Senior Software Engineer / Team Lead · Istanbul, Turkey

- Led hotel-provider integration work and backend services for provider aggregation, search, caching, and data pipelines.
- Designed Go/gRPC services to manage concurrent provider requests and standardise hotel data from external systems.
- Improved reliability through testing practices, Elasticsearch tuning, cache-rate management, and database cluster migration work.
- Guided the team through technical review, testing workshops, and delivery decisions that balanced reliability with business needs.

Hexagonal Architecture · Microservices · Amazon Web Services (AWS) · DevOps · PHP · Go · Object-Oriented Programming (OOP) · Web Applications · Design Patterns · Big Data · Test Driven Development · MySQL · MongoDB · Symfony

Insider

Jun 2019 — Jun 2020

Senior Software Engineer · Istanbul, Turkey

- Refactored newsletter and email-delivery systems from legacy structures toward clearer service boundaries.
- Worked on Go services, messaging, security remediation, and data migration across high-volume marketing data systems.
- Improved maintainability through architecture review, repository-service patterns, and clearer data-flow design.
- Built supporting services for A/B testing, spam detection, and queue-based processing.

RabbitMQ · Amazon Web Services (AWS) · DevOps · PHP · Go (Programming Language) · Object-Oriented Programming (OOP) · Design Patterns · Test Driven Development · MySQL · MongoDB · Leadership · Teamwork · Management · Redis

Bamilo

Aug 2015 — Apr 2019

Senior Software Engineer / Head of Development · Tehran, Iran

- Built and led engineering work across seller tooling, warehouse operations, delivery tracking, inventory services, and internal platforms.
- Owned backend architecture, service design, team growth, and technical review for marketplace operations systems.
- Designed data migration pipelines and service improvements for inventory and seller-centre workflows.
- Helped convert operational problems into maintainable systems for logistics, warehouse, seller, and marketplace teams.

RabbitMQ · Kubernetes · GraphQL · Amazon Web Services (AWS) · DevOps · PHP · Laravel · Go (Programming Language) · Object-Oriented Programming (OOP) · Design Patterns · MySQL · MongoDB · Leadership · Management · Redis · Cassandra · Apache Kafka

Sepehr Mahan Tech

Aug 2012 — Jul 2015

Software Engineer / Senior Software Engineer · Tehran, Iran

- Developed suggestion, task-management, meeting-management, form-builder, workflow, social-network, and mobile API systems.
- Focused on backend services, process automation, maintainable application foundations, and practical delivery for private and public-sector clients.
- Built the early backend and product-engineering foundation that later grew into larger platform, architecture, and leadership responsibilities.

Linux · PHP · JavaScript · Python (Programming Language) · Node.js · SQL · MySQL · Redis · MongoDB · RabbitMQ · Design Patterns · Object-Oriented Programming (OOP) · Teamwork

SELECTED PROJECTS

Pepper.com Platform Work

Marketplace microservices work · Remote / Berlin

Backend services, GraphQL flows, image service work, near-cache behaviour

- Collaborated with Supernova and microservice teams to develop features, maintain services, and improve backend behaviour.
- Redesigned and developed deal submission, product listing, and product detail flows using GraphQL.
- Worked on image-service capabilities, near-cache behaviour, load-balancing improvements, and microservice security review.

GraphQL · Amazon Web Services (AWS) · SQS · SNS · RDS · S3 · ELB · Microservice · PHP · Laravel · Go · Object-Oriented Programming (OOP) · Design Patterns · Test Driven Development (TDD) · ElasticSearch · Redis

RECOMMENDATIONS

Nikola Barac — Head of Software Engineering, Pepper.com

“Kousha worked on pepper.com features as a senior PHP developer and took ownership of one of our most important microservices written in Go. He demonstrated passion for tackling challenging problems, staying o hours until they got solved and following up the next days. While working on his daily tasks he was often paying attention to security issues that many developers often neglect. Kousha was a valuable team member and I would be very happy to work with him again!”

Ryan Klann — VP of Engineering, Pepper.com

“Kousha is a developer who has very strong technical skills and has shown to be able to take responsibility for a project and be relied upon. If there is something technical that he doesn't understand he will jump right in and learn what he needs in order to solve the problem. It's been a pleasure working with you.”

Gizem Sevinç — CTO and Co-founder, Kolay

“Kousha is a talented software developer that I thoroughly enjoy working with. Thanks to its extensive knowledge in backend technologies, it provides effective and fast solutions to many problems. Highly recommended if you are working with PHP, Phyton, or Go.”